

Minguhn Kim

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Education

POSTECH (Pohang University of Science and Technology) Feb. 2025 – Feb. 2027 (Expected)
M.S. in Graduate School of Semiconductor Technology Pohang, South Korea

Laboratory: BEVIL (Broadband Electronics by Virtual Intelligence Lab)

Advisor: Prof. Byungsub Kim

- **Scholarship:** Recipient of the Graduate School Entrance Scholarship.
- **Core Research:** AI-driven EDA: Automated Analog Layout Generation and RL-based Circuit Optimization.
- **Current Focus:** Developing CNN-based layout recognition modules to enhance automation flows.

Chung-Ang University (CAU) Mar. 2019 – Feb. 2025
B.S. in Electrical and Electronic Engineering Seoul, South Korea

- **GPA:** 4.34 / 4.50 (Summa Cum Laude)
- **Military Service:** Sergeant, ROK Army 702nd Special Commando Battalion. (2-years leave of absence)

Key Research & Projects

CNN-based Layout Recognition for Generator Parameter Extraction Aug 2026 – Present
Research Contributor

- Developing a residual CNN framework that classifies cell types and regresses design parameters directly from layout data.
- Building a workflow that converts manual layouts into parameterized generator libraries, leveraging standard and difference datasets for accurate parameter estimation.
- Extending the framework to a previously developed DDI layout generator, exploring the inverse design problem of inferring design parameters from finished layouts.

Samsung S.LSI CMOS Image Sensor (CIS) Layout Optimization May 2026 – Jul 2026
Project Collaborator — RL-based Layout Optimization

- Established a GitHub-based collaboration environment to support the research team's multi-contributor workflow.
- Configured an automated training pipeline on the lab server to enable reinforcement learning (RL)-based layout optimization.

Samsung AI Cluster Project 2025 – 2026
Lead Developer — DDI Layout Automation

- Engineered a GUI-based routing automation tool for Display Driver IC (DDI) layout, streamlining industrial design workflows.
- Utilized AI-native development environments (Claude Code) to accelerate software architecture design and algorithm prototyping.
- Implemented A* search algorithms to achieve robust automation in analog circuit design.

CTLE Optimization via Reinforcement Learning 2025
Research Contributor

- Contributed to a Python-based framework designed to automate Continuous-Time Linear Equalizer (CTLE) parameter tuning.
- Applied Actor-Critic algorithms to bridge the gap between complex analog behaviors and automated design flows.

Technical Skills

- **AI & Programming:** Python (PyTorch, NumPy, Pandas), Claude Code, C/C++, Git.

- **Design Tools:** Cadence Virtuoso (Custom IC Design/Layout).

Awards & Languages

- **Dean's Award (College of ICT Engineering):** 2024 CAU Engineering Academic Festival Dec. 2024
- **Academic Excellence Scholarship:** Chung-Ang University (Multiple-semester recipient)
- **Languages:** OPIc Advanced Low (AL) (Jan. 2026), NewTEPS 387 (Mar. 2024)

Extracurricular Activities

- **Music Production:** Singer-songwriter with proficiency in piano, guitar and DAWs; focuses on creative problem-solving through musical composition.
- **Physical Fitness:** Maintains disciplined endurance through consistent weight training and running.